Evaluating Anti-Virus Effectiveness in Linux

G. Raffa, D. Sgandurra, D. O'Keeffe
Royal Holloway, University of London
IEEE International Conference on Big Data 2022
BDACCIP Workshop

Introduction

- **Why evaluating Linux Anti-Virus software?**
  - Linux is not malware-free
  - Cross-platform threats (e.g., JavaScript)
  - Windows Subsystem for Linux (WSL)

- **What do we know about Linux malware?**
  - Recent studies show that malware targeting IoT devices is growing in complexity (e.g., crypto-mining)

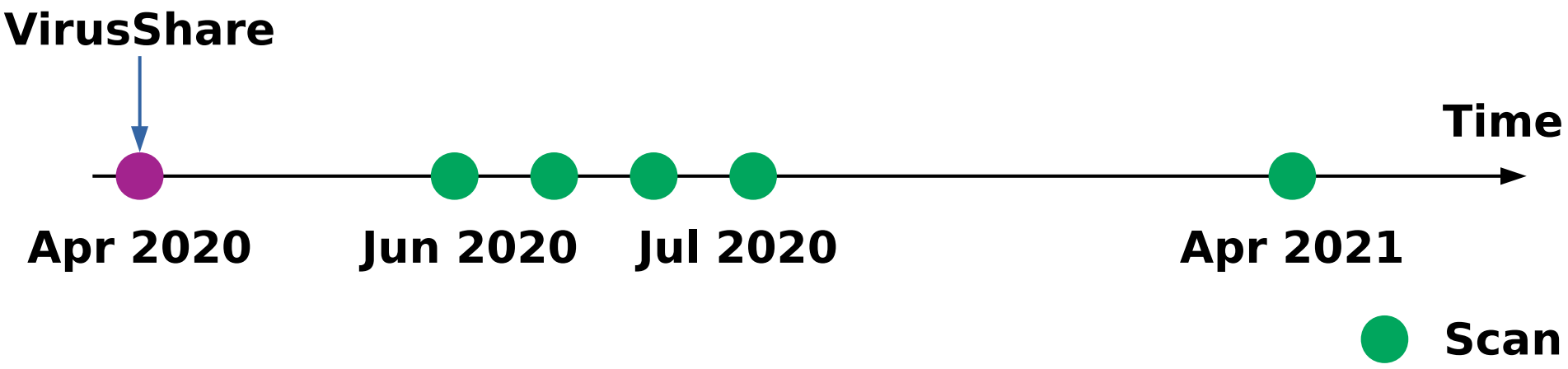
Despite this threat landscape, very little information about the effectiveness of Linux AVs is available

Research Objectives

How effective are Linux AVs (detection rate)?	→	Locally-installed and online AVs (VirusTotal)
Do Linux AVs show any regression effects?	→	Multiple tests over a period of ten months
What are the capabilities of the undetected samples?	→	Open-source capability analysis tool (CAPA)
Any difference among local and online versions?	→	Detection rates and regression compared

Tests with Virtual Machines (1)

- **Methodology**
  - Malware repository (43,553 ELF samples)
  - Four locally-installed AV programs (widely used)
  - Each AV installed in a dedicated virtual machine
  - Four scans run over the course of three weeks
  - Fifth scan after ten months



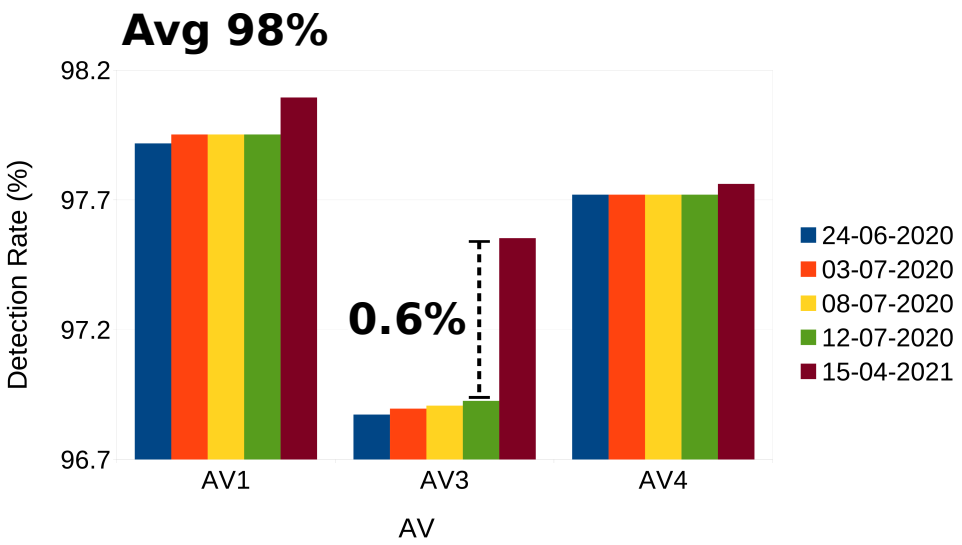
Tests with Virtual Machines (2)

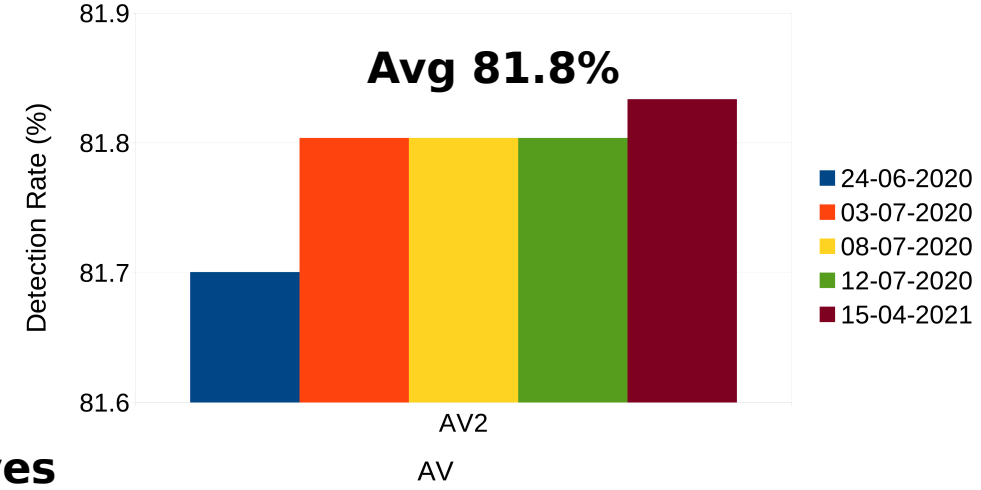
- **Detection rate**
  - Definition:

$$DR = \frac{TP}{TP + FN} \quad (*)$$

  - Results of last scan show marginal improvements:

Largest detection rate increase in comparison to second last test is 0.6% (AV3)



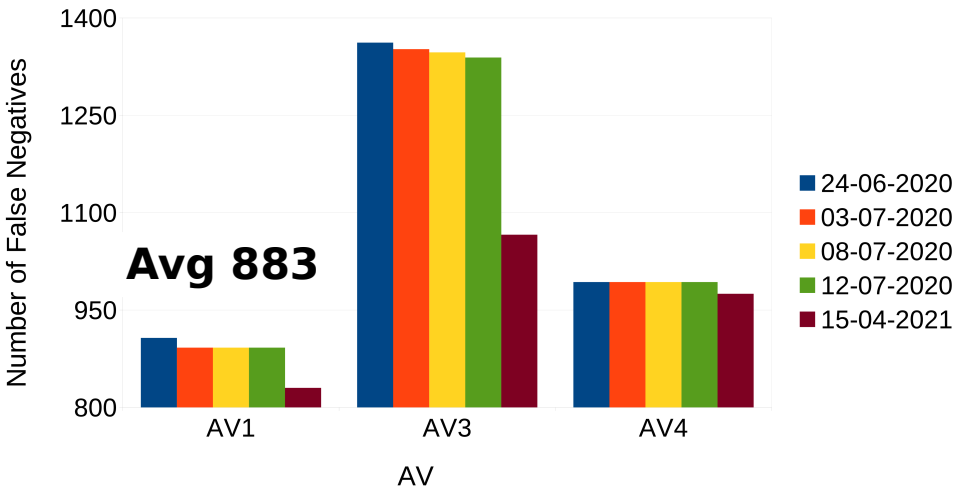


(*) TP = True Positives / FN = False Negatives

Tests with Virtual Machines (3)

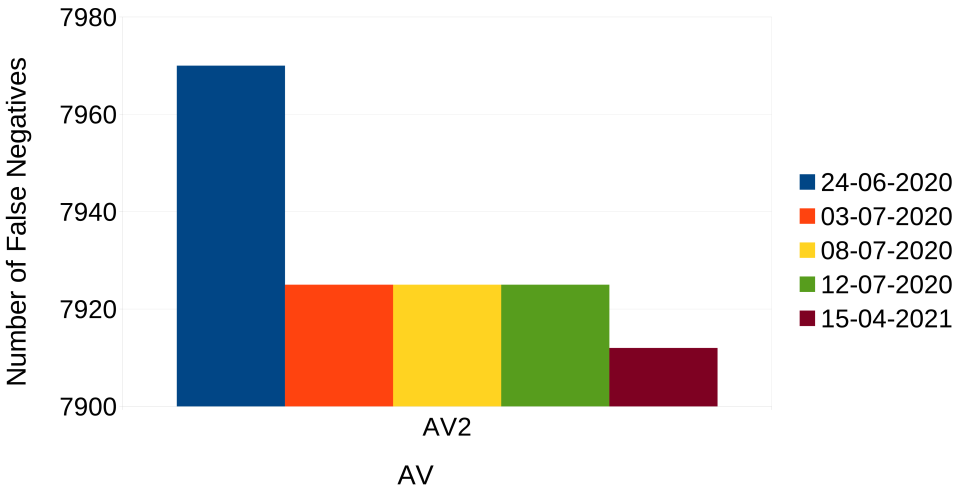
- **False negatives**
  - For AV1 (highest DR), 883 false negatives recorded (avg)

Surprisingly high number



- **Regression effects**
  - Results post-processed on a per-file basis

AV1 and AV4 affected by regression (6 and 3 samples)



Malware Capability Analysis (1)

- **CAPA Tool**
  - Identifies capabilities by using rules
  - Supports only 32 and 64-bit x86 architectures
  - Includes ELF-specific rules (36 in total)
- **Methodology**
  - Created two subsets

False negatives

↓

Malicious files never flagged in our local tests (8,337)

True positives

↓

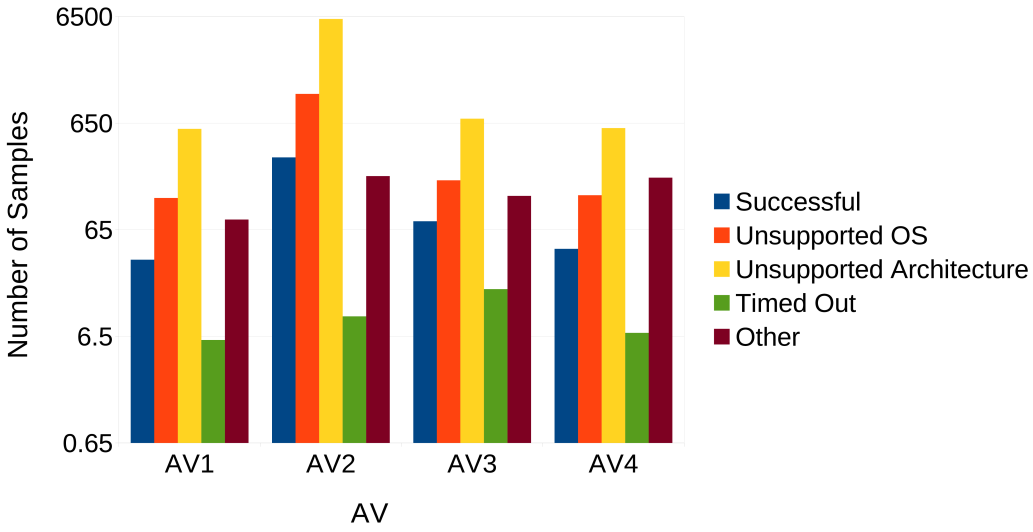
Malicious files detected at least once in our local tests (42,962)

Malware Capability Analysis (2)

- **False negatives**
  - 313 samples processed
  - Why only 3.8%?

    Unsupported architecture

    Unsupported OS



- **Results**

High-level Capability
host-interaction
anti-analysis

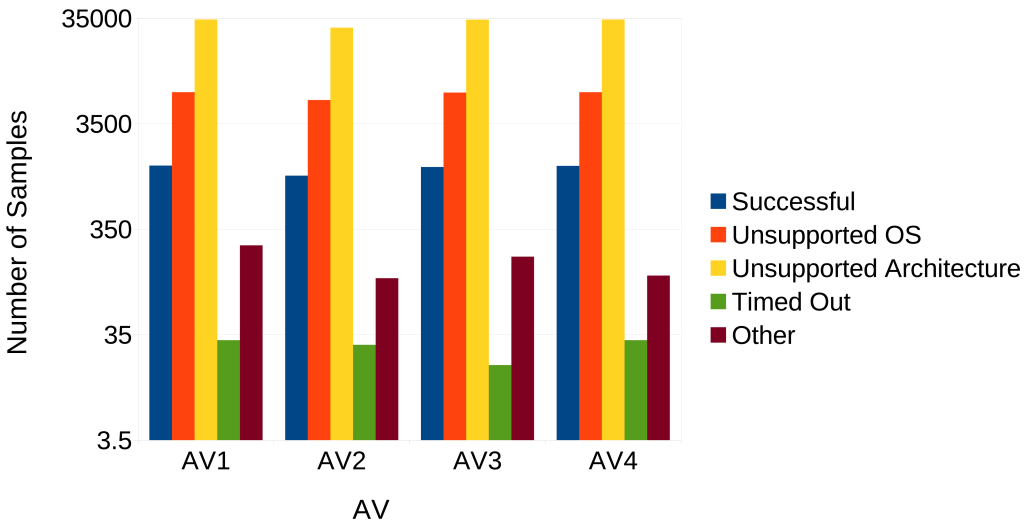
CAPA Namespace
anti-analysis/packer/upx
anti-analysis/anti-vm/vm-detection

Malware Capability Analysis (3)

- **True positives**
  - 1,273 samples processed
  - Why only 3%?

    Unsupported architecture

    Unsupported OS



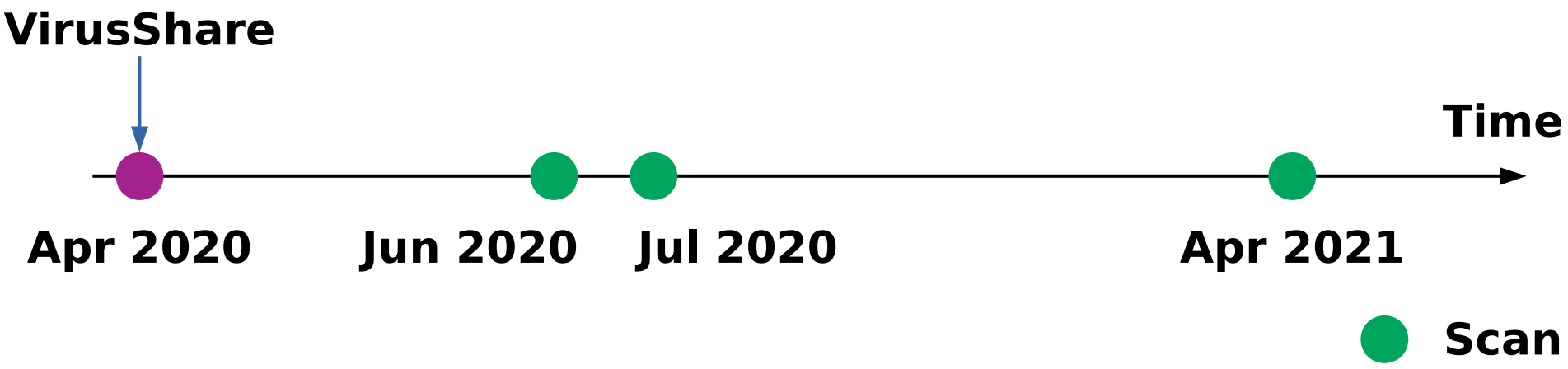
- **Results**

High-level Capability
anti-analysis
data-manipulation

CAPA Namespace
anti-analysis/packer/upx
anti-analysis/anti-vm/vm-detection

Online Tests with VirusTotal (1)

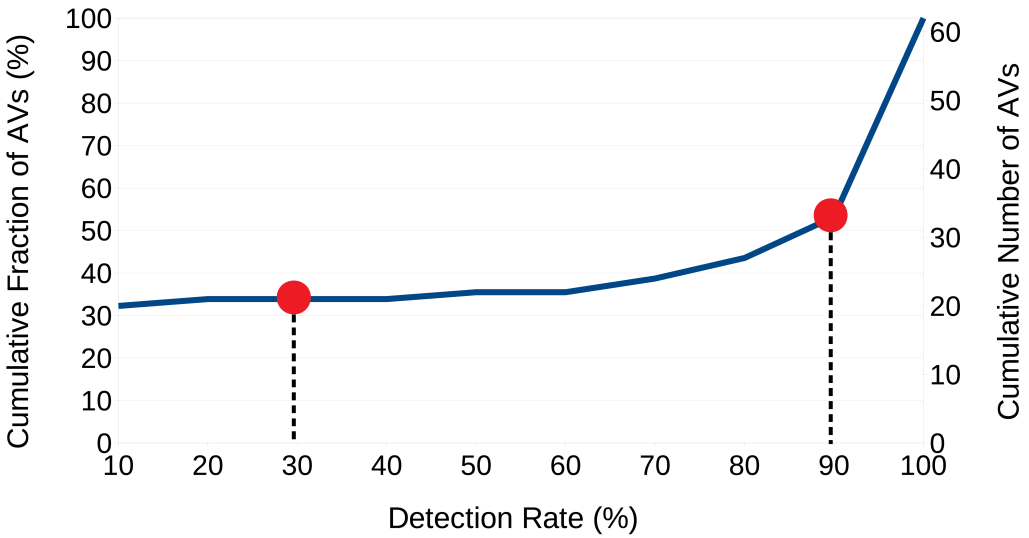
- **Methodology**
  - 4,000 samples from same dataset randomly selected
  - Samples submitted via VirusTotal API
  - Two scans run over the course of two weeks
  - Third scan after ten months



Online Tests with VirusTotal (2)

- **Detection rate**
  - Average DRs of the online AVs significantly lower than those of the locally-installed ones

Overall avg DR 61.3%



  - Cumulative distribution

DR is at most 30% for one third of the online AVs

Only 50% of the online AVs have a DR above 90%

Online Tests with VirusTotal (3)

- **Regression effects**
  - Total amount of flagged files decreases for some AVs

Test 1 Vs Test 2
13 AVs affected

Test 2 Vs Test 3
24 AVs affected

- **Online AVs Vs Locally-installed AVs**
  - Only small discrepancies found

AV	Delta (%)
AV1	-2.2
AV2	-1.2
AV3	-0.6
AV4	0.1

Unlike similar figures available in the literature, these are Linux-specific

Conclusion

How effective are Linux AVs (detection rate)?	→	Detection rates increase very marginally over time
Do Linux AVs show any regression effects?	→	Found evidence of overall and file-level regression
What are the capabilities of the undetected samples?	→	No major difference between FN and TP
Any difference among local and online versions?	→	Only minor, but we provide Linux-specific figures

Thank You!

- **Analysis framework**

  https://github.com/giusepperaffa/capa-tool-launcher

- **Any questions?**

  giuseppe.raffa.2018@live.rhul.ac.uk